Case Study on the Demonstration of Network Communication Models (OSI/TCP-IP) Using a Network Simulation Tool

Aim:

To demonstrate the OSI and TCP/IP network communication models using a network simulation tool (Cisco Packet Tracer) and to analyze how data moves across the layers of a real-world network.

Objective:

- Study the OSI and TCP/IP network communication models.
- Build a network topology using a network simulation tool.
- Observe how data is encapsulated and de-encapsulated at each layer.
- Compare the OSI model with the TCP/IP model.
- Verify end-to-end communication using standard protocols and test cases.

Problem Statement:

Understanding network communication models only through theoretical study can be difficult for beginners. Students may not clearly understand how data is transmitted, how IP and MAC addresses are used, or how network protocols support communication.

Therefore, a network simulation tool is required to demonstrate the communication process practically and provide a clear understanding of the OSI and TCP/IP models.

Network Simulation Tool

Cisco Packet Tracer:

Cisco Packet Tracer is a network simulation application developed by Cisco. It allows users to create virtual network topologies, configure network devices, and test communication between connected systems.

Features:

- Provides virtual PCs, switches, routers, and other network devices.
- Supports IP address configuration.
- Allows users to test connectivity using commands such as ping.
- Provides Realtime Mode and Simulation Mode.
- Displays network packet movement and protocol information.
- Helps identify and troubleshoot network configuration errors.

Hardware and Software Requirements:

1. Cisco Packet Tracer software
2. Two end devices (PCs).
3. One Cisco 2960 switch.
4. Copper straight-through Ethernet cables.
5. IP addresses and subnet masks for the PCs.

Network Topology:

The network consists of two computers connected to a central switch. The switch enables communication between the computers within the same local area network.

PC0 → Switch → PC1

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Both PCs belong to the same IP subnet, so a router is not required for communication between them.

Procedure:

1. Open Cisco Packet Tracer.
2. Place two PCs and one Cisco 2960 switch in the workspace.
3. Connect PC0 to the switch using a copper straight-through cable.
4. Connect PC1 to the switch using another copper straight-through cable.
5. Open PC0 and select **Desktop → IP Configuration**.
6. Enter IP address 192.168.1.1 and subnet mask 255.255.255.0.
7. Configure PC1 with IP address 192.168.1.2 and the same subnet mask.
8. Wait for the switch connections to become active.
9. Open the command prompt on PC0.
10. Enter `ping 192.168.1.2` and observe the response.

Demonstration of the OSI Model:

The Open Systems Interconnection (OSI) model is a conceptual framework that explains how data communication takes place between network devices. It contains seven layers, and each layer performs a specific function.

1. **Application Layer:** The Application Layer provides network services to user applications. Examples include web browsing, email, and file transfer. Protocols such as HTTP and DNS operate at this layer.
2. **Presentation Layer:** The Presentation Layer deals with data representation, encoding, encryption, and compression where required. It ensures that information can be interpreted correctly by the receiving application.
3. **Session Layer:** The Session Layer establishes, manages, and terminates communication sessions between applications.
4. **Transport Layer:** The Transport Layer supports communication between application processes. TCP provides reliable, ordered delivery, while UDP provides connectionless transport. ICMP ping messages do not use TCP or UDP.
5. **Network Layer:** The Network Layer handles logical addressing and packet delivery across networks. IP addresses identify the source and destination devices at the network layer.
6. **Data Link Layer:** The Data Link Layer organizes data into frames and uses MAC addresses for delivery over a local Ethernet network. The switch examines destination MAC addresses to forward frames.
7. **Physical Layer:** The Physical Layer transmits bits through the network medium. In this experiment, Ethernet cables provide the physical connection between the PCs and the switch.

Data Transmission in the OSI Model:

When PC0 sends a ping request to PC1, the data is processed through the networking stack. IP and ICMP information is carried inside an Ethernet frame. The switch forwards the frame toward the destination. The receiving PC processes the incoming information and sends an ICMP echo reply if communication is successful. This process demonstrates the general concept of encapsulation, in which each relevant protocol layer adds its own control information before transmission.

Demonstration of the TCP/IP Model:

The TCP/IP model is used to explain communication across modern computer networks, including the Internet. It consists of four layers.

1. **Application Layer:** This layer provides network services to applications. It combines the functions associated with the Application, Presentation, and Session layers of the OSI model. Examples: HTTP, DNS, and FTP.
2. **Transport Layer:** This layer provides communication between applications running on different devices. **TCP:** Provides reliable and ordered data delivery. **UDP:** Provides connectionless data delivery with lower protocol overhead.
3. **Internet Layer:** This layer handles IP addressing and packet delivery between devices and networks. ICMP is used for diagnostic functions such as ping.
4. **Network Access Layer:** This layer deals with local network communication, Ethernet frames, MAC addressing, and transmission over the network medium.

Simulation Observation:

In Cisco Packet Tracer, select Simulation Mode and generate a ping packet. Use Capture/Forward to observe the packet movement. The Event List and packet details can show protocols such as ARP and ICMP, depending on the current stage of communication and the simulation state.

Conclusion:

This case study demonstrates network communication models using Cisco Packet Tracer. By connecting two PCs through a switch, configuring IP addresses, and testing connectivity with the ping command, the basic process of data transmission can be understood.

The study explains how the OSI model divides network communication into seven layers and how the TCP/IP model organizes communication into four layers. Simulation Mode provides a useful way to observe packet movement and examine protocols such as Ethernet, IP, ARP, and ICMP.

Thus, Cisco Packet Tracer serves as an effective educational tool for learning network communication, understanding protocol functions, and developing basic network troubleshooting skills.